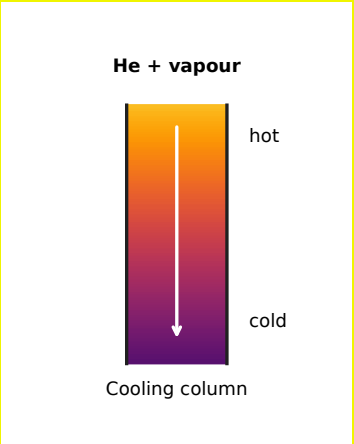
PSI Center for Scientific Computing, Theory and Data

GROUP MEETING

LESTO project update

From Bojan’s baseline to conservative Pb–Bi–I deposition



Jan Kren
Paul Scherrer Institute | 15 October 2026

Project status

The coupled model is implemented; predictive validation remains open.

9
Reactive gases
Pb–Bi–I; helium carrier

4
Wall inventories
PbI_2, BiI_3, Pb, Bi

19
Production reruns
All pass numerical gates

64
Follow-up cases
Completed input sensitivities

41,472
Production cells
Bounded limited-linear advection

18 K
Onset RMS error
5 independent PbI_2 experiments

Snapshot: 14 October 2026. Production = 15 baseline cases + 4 silica mappings; 64 follow-ups use the earlier upwind mesh.

Bojan’s baseline

I retained the conservative mock and added a separate physical phase-change path.

Component	Bojan’s foundation	My additions
Transport	Frozen-flow species solver	Paired gas–wall exchange
Phase change	PbI_2 gas-to-solid mock	Reversible, inventory-limited HKS law
Conservation	Realised gas sink feeds solid	Gas, wall and Pb/Bi/I element ledgers
Thermochemistry	xGEMS probe / startup demo	Optional GEMS bridge + gas equilibrium
Evidence	Evolutionary pipe-flow cases	Regression, refinement, experiments

Mock path preserved; physical runs use the specialised gas kernel.

Attribution: phase-change plan, section 31; baseline tag 0.1 (7155e6b), integrated main 71bf192.

Reversible wall exchange

Wall kinetics and radial diffusion now share one conservative exchange flux.



$$j_i = G_i^{\mathrm{eff}} \left[p_{\mathrm{eq},i}(T_w) - p_{i,c} \right]$$

$$\frac{1}{G_i^{\mathrm{eff}}} = \underbrace{\frac{1}{G_i}}_{\text{kinetics}} + \underbrace{\frac{\chi_i d_f}{(\rho D_i)_f}}_{\text{half-cell diffusion}}$$

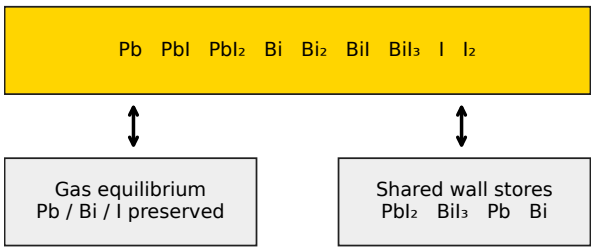
- Deposition and re-evaporation
- No evaporation from an empty store
- Equal-and-opposite gas/wall transfer

No prescribed deposition temperature in the HKS model.

$j_i > 0$: wall to gas; $\chi_i = \rho RT/M_i$; d_f: cell centre to wall. HKS = Hertz–Knudsen–Schrage.

Multispecies thermochemistry

Chemical redistribution now conserves Pb, Bi and I across shared wall stores.



- 9 gases; 4 condensed inventories
- Bi and Bi_2 share one Bi store
- Optional frozen/local GEMS modes

Reported column runs: ideal-gas kernel.

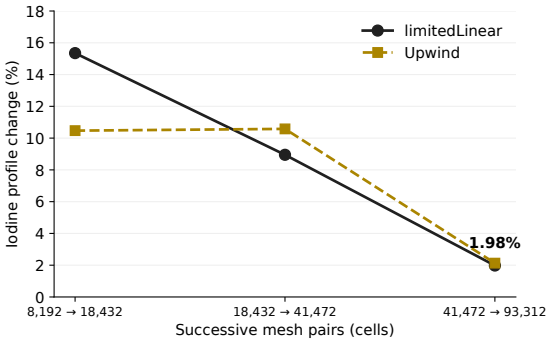
Frozen helium; no thermal feedback.

$$3\,\mathrm{BiI(g)} \longrightarrow 2\,\mathrm{Bi(cond.)} + \mathrm{BiI_3(g)}$$

Implemented reactive channel is irreversible; equilibrium is a model assumption, not measured reaction kinetics.

Numerical verification

Spatial transport error mattered more than time step or corrector count.



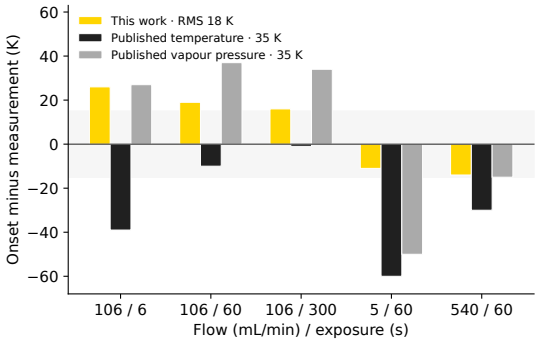
- 41,472 cells; limitedLinear
- Finest mesh change: 1.98%
- Raw element residue: 1.41×10^{-12}

1.98% is a successive-mesh difference, not a certified error bound.

Shifted-Bil steel case; $E_{L1} = \sum_b |I_b - I_b^{\mathrm{ref}}| / \sum_b I_b^{\mathrm{ref}}$. No formal grid-convergence index.

Published PbI_2 benchmark

Onset-temperature RMS error is 18 K, compared with 35 K for both published models.



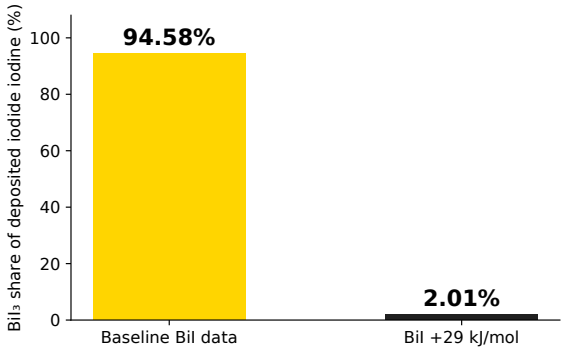
- 5 experiments; no wall-law retuning
- Deposit peaks 1–4 cm upstream
- 3/5 errors exceed ± 15 K

Source mole fraction reaches 0.33–0.36: passive-carrier assumption fails locally.

Lobresco et al. (2026), Table 2; run/035. Comparison to published model results, not a new T-Flows run.

Thermodynamic sensitivity

The BiI thermodynamic assumption can reverse the predicted iodide partition.



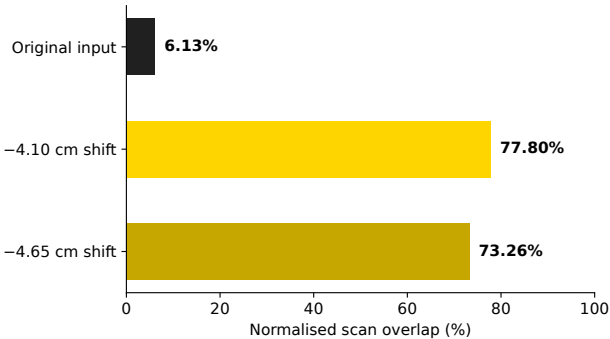
- Steel Q2; 1% metal saturation
- BiI_3 iodine share: 94.58% to 2.01%
- Separation survives refinement

$+29$ kJ/mol is a surrogate perturbation, not a measured correction.

$f_{\mathrm{BiI_3}} = 3N_{\mathrm{BiI_3}}/(2N_{\mathrm{PbI_2}} + 3N_{\mathrm{BiI_3}})$: deposited iodides only. Q2 iodine release remains prescribed.

Silica temperature registration

A 4–5 cm temperature-coordinate discrepancy dominates the original silica mismatch.



- Original Q1 peaks: 56 / 63 cm
- Measured peaks: 51 / 59 cm
- Shift hypothesis: 51 / 59 cm

Both mappings are unconfirmed. Their overlap ranking reverses with the advection scheme.

Liu et al. (2025); Q1 production cases. Shift temperature only; keep measured scan fixed. No CFD-fitted alignment.

Next steps

The method is ready for review; better inputs are needed for predictive validation.

Completed

- Conservative physical extension of the baseline
- Numerical and input-sensitivity evidence
- Manuscript and D3.8 draft

Next evidence

- Original temperature / scan coordinates
- Source history and BiI thermodynamics
- Cooling protocol; broader spatial checks

For discussion

Which input uncertainty comes first? What is needed for manuscript sign-off?